FINITE GENERATION OF A MODULE IS DETECTED AFTER A FINITE PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A finite morphism is affine. Morphisms, the standard properties of finite morphisms (affineness). Reduces the sheaf statement to a statement about a finite ring map.
- A module over a finite extension ring is finite if and only if it is finite over the base. Algebra, Lemma algebra-lemma-finite-module-over-finite-extension, invoked at proof lines 2307-2309. Gives the equivalence for the B-module M.
- Finite type of a quasi-coherent module is detected by its associated module on an affine scheme. Properties, Lemma properties-lemma-finite-type-module, invoked at proof lines 2310-2312. Converts between the algebra statement and the sheaf statement.

Source reference. Original lemma 00GJ: sources/stacks-chow/tags.txt, source lines 609–609. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01PB: sources/stacks-chow/tags.txt, source lines 2224–2224. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Assume f is a finite morphism. Then $\mathcal{F}$ is an $\mathcal{O}_X$-module of finite type if and only if $f_*\mathcal{F}$ is an $\mathcal{O}_Y$-module of finite type.*

Proof. As f is finite it is affine. This reduces us to the case where f is the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ given by a finite ring map $A \to B$. Moreover, then $\mathcal{F} = \widetilde{M}$ is the sheaf of modules associated to the B-module M. Note that M is finite as

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

a B-module if and only if M is finite as an A-module, see Algebra, Lemma 00GJ. Combined with Properties, Lemma 01PB this proves the lemma. $\square$